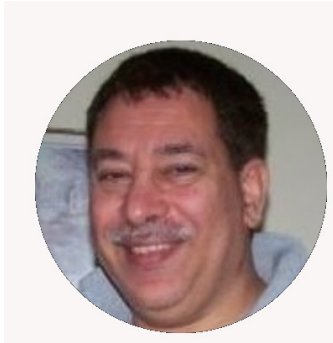


Nicholas (Nico) Michael

| Senior Java Developer Team Lead | Solutions Architect



1. Personal Information

Nationality: South African

Location: Pretoria, Gauteng, South Africa

Languages: English & Afrikaans

Contact Details

Email: nicomichael2018@yahoo.com

Cell: 082 508 1950

YouTube DevOps Channel : <https://www.youtube.com/@dvp7388>

GitHub: <https://github.com/nic0michael>

AI Portfolio video: https://youtu.be/GyjAqyJZw_Y

Portfolio Development project using Outbox and VETO Design patterns:

<https://github.com/nic0michael/Trans-Process-Platf/tree/master>

Professional Summary

Senior Java Developer, Technical Lead and Solutions Architect with **26+ years of Java** development experience and **6 years of architecture experience** across Infrastructure and Solution Architecture. 5 years of Python coding.

Delivered 56+ Spring Boot microservices into production at WesBank

I worked across Banking, Finance, Healthcare, Insurance, Telecommunications, Education, Loyalty and Utility sectors.

Strong practical experience in **Java, Spring Boot, Kubernetes, Docker, Terraform, Azure AKS** and CI/CD.

Development Best Practice Coach specializing in TDD, BDD, Clean Code, DevOps and AI-assisted development.

How do I approach a problem?

I try to understand the problem before jumping to a solution. I ask questions to get the right context, research the problem, consider different solutions and their trade-offs, and then implement and validate the chosen solution.

An example from my time at Cell C:

When I joined Cell C, I found that the Synix reporting system was very slow. The reports took around 30 minutes for all to run because they performed **expensive queries** with many joins against the production database of the cellular system.

I recognized that this was not only a performance problem. Running these queries against the production database could potentially **affect the system supporting customer calls** on the Cellular System.

I redesigned the solution by **introducing a staging database, isolating the reporting workload** from the production system. I also created dedicated reporting tables so that reports did not have to repeatedly execute expensive queries across multiple databases.

The result was immediate: reports that previously took around **30 minutes were reduced to approximately 3 minutes**, while removing the reporting workload from the production system.

There was another architectural problem. **The existing system had been designed for Siemens/Nokia base stations**, but Cell C had started installing Huawei equipment.

Rather than modifying the existing code specifically for Huawei, I changed the design so that the **data-capture components implemented a common interface** and used a Factory Design Pattern. This allowed separate implementations for Siemens/Nokia and Huawei while keeping the rest of the system independent of the underlying technology.

This also meant that future base-station technologies could be introduced without having to redesign the entire system.

This is typical of how I approach problems: I look beyond the immediate issue and try to find a solution that improves the system while also making it more maintainable and adaptable for the future.

Core Competencies

- Software Engineering: **Java, Spring Boot, REST APIs, JPA, Full Stack Development**
- Front-end Development: **Angular 20, Cypress**(similar to Playwright), Jasmine
- Technical Leadership: **Team Leadership, Mentoring, Training, Development Best Practices**
- DevOps & Cloud: **Kubernetes, Docker, Helm, Jenkins, GitLab CI/CD, Terraform**
- Cloud Platforms: **Azure AKS, AWS, GCP, Oracle Cloud**
- Architecture: Solution Architecture, Infrastructure Architecture, Enterprise Architecture
- Cloud-Native: **Kubernetes, K3S, AKS, Docker, Helm**, Cloud-Native Architecture
- Software Quality: TDD, BDD, **JUnit, Mockito, SonarQube**, Clean Code
- REST: Microservices **Quarkus 3.8, Spring Boot**, and Python
- Database: **MMSQL, PostgreSQL, Oracle 11g to 19c, MySQL**, DB2
- JMS Queues: **RabbitMQ, Kafka**, IBM MQ series
- AI-Assisted Development: Spec Driven Development SDD, Context Driven Development CDD, Prompt Engineering, Multi-Agent Development
- AI Tools: **GitHub Copilot, Claude Code, Codex, Cursor**
- Process & Architecture: **UML, C4, TOGAF, IBM GS Method**
- SASS: **The Billing switch** was delivered as a SASS in **Azure AKS Kubernetes**

DevOps & Cloud Engineering Experience

Practical experience across DevOps, cloud infrastructure, automation, containerization, Infrastructure as Code, CI/CD and Kubernetes.

- 2 years **Azure** experience including **Azure AKS Kubernetes** and CI/CD Pipelines.
- 1 year **GCP** experience developing a Notification App using Firebase in GCP.
- 1 year practical **AWS** experience building infrastructure for Udemy training.
- 4 years **Oracle Cloud** provisioning **Rocky Linux** servers
- 8 years **Docker** experience.
- 7 years **Kubernetes, including Azure AKS Kubernetes** and Cloud-Native platform experience.
- Applied **Terraform** for repeatable Infrastructure as Code deployments.
- Developed PowerShell and Linux shell automation for development, testing, analysis and deployment.
- Worked with **Docker Compose and multi-container environments**.
- Practical Kubernetes experience covering K3S, workloads, nodes, namespaces, pods, services and troubleshooting.
- Experience with Proxmox virtualised infrastructure.
- Experience with **SonarQube** for automated code-quality and security analysis.
- Practical troubleshooting of networking, ports, DNS, certificates, containers, virtual machines and deployment failures.
- Applied **DevSecOps** principles within the software delivery life-cycle.

Architecture Experience

Infrastructure Architect: IBM, ESKOM – 3 years

Solution Architect: Cell C, FleetDomain, Vertice – 3 years

Designed and supported enterprise, infrastructure and application architectures across financial services, telecommunications, utilities and healthcare environments.

Implemented an Architectural Decision Register

Created an Architectural Methodology based on TOGAF, C4, and IBM GS Method.

AI-Assisted Development & Engineering Leadership

Led research, evaluation and adoption of AI-assisted software development tools including GitHub Copilot, Claude Code, OpenAI Codex and Cursor.

- Researched and implemented Spec-Driven Development using Microsoft GitHub Spec Kit.
- Developed corporate AI policy files and coding governance guidelines for AI-generated code.
- Evaluated single-agent and multi-agent development approaches.
- Evaluated competing AI platforms using a common full-stack application and recommended Claude Code and Codex based on practical results.
- Researched local LLM development using Ollama, OpenCode and Google Gemma.
- Created the Iterative Context-Driven Development (ICDD) methodology using AI-assisted brainstorming, requirement refinement and iterative quality reviews.
- Produced technical training content through a DevOps-focused YouTube channel.
- Using **RAG files** and **Claude Code, Context Driven Development** I was able to reverse engineer and **Document the project Architecture** of our projects
- I generated the following project using **Python with FastAPI and CORS** and an Angular front-end
<https://github.com/nic0michael/ClaudeCodeMultiAgent>

Key Achievements

- Delivered **56 Spring Boot microservices** into production at WesBank.
- Designed and implemented a **High Availability K3S Kubernetes cluster** spanning Germany, England and Johannesburg.
- **Developed a Python-based K3S Kubernetes Cluster Generator** to automate Kubernetes cluster configuration and generation.
- Used Helm Charts to automate application deployment and management across Kubernetes.
- Created technical demonstrations **integrating Terraform, Kubernetes, Helm and Proxmox**.
- Introduced TDD, BDD and DevOps best practices across development teams.
- Developed the AHAA Agentic Development Methodology.

- **Lead the integration of the Billing Switch** application into **Azure** AKS Kubernetes.
- Published a Microservices White Paper.
- Published 50 Open Source projects on GitHub, including K3S Auto Cluster Generator and Claude Code Multi Agent With Policies.
- **Provided technical training** in Java, Kubernetes, DevOps and AI-Assisted Development.

Key Projects

WesBank: Credit Scoring System 2018

Role: Senior Java Developer Team Lead

I was hired to deliver a Credit Scoring System, I **analyzed requirements** and designed the solution using UML. Designed a **non-synchronous, parallel-processing architecture** using RabbitMQ across a 4-node cluster, with Oracle for credit-score results and a micro-service for end-user access. Researched and prototyped a Drools rules engine to manage extensive business rules. Led implementation through to Production; The scoring service code remained unchanged for 3 years

IBM: System Architecture for GJMC Metro Council of Johannesburg 2001

Role: System Architect

Took responsibility for the software and networking architecture across the GJMC Metro Council of Johannesburg. The requirement was to develop a complete Systems Architecture covering all customer regions, each with its own existing systems and requirements.

I visited each region to **identify and understand the existing systems** and infrastructure. I then produced a complete set of Architecture artefacts for each region, together with the overall GJMC Systems Architecture and the supporting documentation. This provided a consolidated architectural view of a large, geographically distributed environment

IBM: Architecture deliver for StanLib development project 2002

IBM had been contracted by StanLib to deliver an Insurance System **based on IBM's Insurance UML Model**, with development and implementation to be performed by a subcontractor. My responsibility was to ensure that the proposed system had the required infrastructure architecture to support its development and implementation.

I analyzed the proposed system and created the complete Infrastructure Architecture artefacts and supporting documentation required by the subcontractor. My colleague was responsible for the Solution Architecture; together we delivered the complete solution, including the required infrastructure, to the subcontractor for development and implementation.

Vertice: Dockerizing the Billing Switch Project taking it into the Azure AKS Kubernetes cluster 2025

This was done in 5 phases:

1. **Dockerized the Billing Switch** so new hires could easily install the required applications on their laptops.
2. **Taught Kubernetes to the team** performing the Azure AKS POC, including the Team Leader and DevOps team.
3. **Led the POC** team, creating the Docker container and Kubernetes manifests.
4. **I created the Docker Containers** in our Azure DevOps Repository
5. Worked with DevOps, we created the pipelines and deploy the Billing Switch to the Development environment.
6. **I resolved Issues with the Kubernetes Manifests**, then deployed the Billing Switch to UAT and ultimately to Production.

Employment History

Vertice

Solution Architect | Aug 2025 – Present

Focus: Enterprise Solution Architecture, AI Agentic Systems R&D, AI Architecture, Agent-Oriented Design, Technical Strategy and Innovation.

- Maintained the Architecture Intellectual Capital repository in SharePoint.
- Maintained the Architectural Decision Register (ADR) process and repository.
- Maintained the System Architecture Review (SAR) framework and documentation.
- Researched and evaluated emerging AI-assisted software development tools.
- Researched Spec-Driven Development using GitHub Spec Kit.
- Researched Claude Code, Cursor and Codex.
- Investigated single-agent and multi-agent AI development approaches.
- Developed corporate AI policy frameworks and coding governance guidelines.
- Created and maintained the AHAA methodology, drawing on C4, TOGAF, Zachman and IBM Global Services methodologies.
- Developed Tech-Radar applications providing real-time technology adoption information for management.
- Developed applications providing management with Code Quality, Security Issues and Technical Debt information, project managers with Git repository status, and business analysts with development task status.
- Provided training on Agentic Development tools.

Tech Stack: **GitLab**, Azure DevOps Repo, Azure AKS Kubernetes, GitHub Copilot, GitHub Spec Kit, Claude Code, Cursor, Codex, VS Code, IntelliJ, Java, Python, Angular, Spring Boot **Python, Bash, Power Shell.**

Vertice

Senior Java Developer | Full Stack Developer | Best Practice Coach | Aug 2024 – Aug 2025

- Developed application code and provided technical training to development teams.
- Led the Dockerization of the Billing Switch.
- Mentored team members in Docker adoption.
- Participated in the Azure AKS Kubernetes POC.
- Guided the team through its first AKS Kubernetes adoption and integration.
- Provided Kubernetes and cloud-native development training.
- Introduced and delivered Helm training to the DevOps team.
- Trained development teams in TDD.
- Introduced and trained teams on Spec-Driven Development.
- Taught Context-Driven Development using Claude Code.
- Introduced advanced brainstorming techniques to improve SDD and CDD outcomes.

Tech Stack: Java, Spring Boot, Docker, Dockge, Kubernetes, K3S, Azure AKS, Helm, **GitLab**, Azure DevOps, **Python, Bash, Power Shell, Prometheus**

LoyaltyPlus

Senior DevOps Consultant / Senior Developer | Jun 2023 – Jul 2024

- Designed and **implemented Disaster Recovery** and backup strategy for customer WordPress platforms using Ansible, Docker, HAProxy and Semaphore.
- Built and automated **High Availability K3S Kubernetes cluster** across Germany, England and South Africa.
- Developed **infrastructure automation using Ansible, Terraform, Proxmox**, Cockpit and Rocky Linux.
- Developed a distributed messaging platform using Camel Quarkus, Kafka, Angular and REST APIs supporting SMS, Email, WhatsApp, Push Notifications and In-App Notifications.
- Implemented Firebase integration for mobile notification services.
- Designed security monitoring and vulnerability scanning using Metasploit and StarRocks.
- Evaluated and implemented Zero Trust networking using NetBird.

Tech Stack: Java, Spring Boot, **Camel, Quarkus, Kafka, Angular, Docker, K3S Kubernetes, Helm, Ansible, Terraform, Proxmox, HAProxy, Semaphore, NetBird, Metasploit, Python, BASH, StarRocks, Google GCP, Firebase, REST APIs, Rocky Linux Servers.**

Advance / MTN

Senior Software Engineer | Kubernetes Specialist | Technical Trainer | Jun 2022 – Jun 2023

- Developed solutions for the MTN USSD platform.
- Designed and implemented **High Availability K3S Kubernetes clusters** on Red Hat infrastructure.
- Created Kubernetes manifests and Helm Charts for Zookeeper, Elasticsearch, Kibana, Prometheus, PostgreSQL and MongoDB.
- Led Kubernetes adoption and deployment practices across development teams.
- Delivered training in TDD, JUnit, Mockito, Kubernetes, Helm and cloud-native deployment practices.

Tech Stack: Java, Spring Boot, **Kubernetes**, K3S, **Helm**, **Docker**, Elasticsearch, Kibana, **Prometheus**, PostgreSQL, MongoDB, Zookeeper, JUnit, Mockito, **Jenkins**, **Terraform**, **Bash**, **Python**

DVT

Senior Java Developer Team Lead | Feb 2017 – Jun 2022

Discovery | 2022

- Led Spring Boot Micro-service development initiatives.
- Developed Java training curricula and technical enablement programs.

Tech Stack: Java, Spring Boot Micro-services, Hibernate, GitLab, TDD, Mockito.

Curro Schools | 2021

- Served as Software Development Best Practices Coach and Senior Java Developer.
- Delivered Proofs of Concept covering TDD, BDD, **Docker**, **SonarQube**, Portainer, RabbitMQ, Postman and Micro-service architecture.
- Trained development and QA teams in automated testing and engineering best practices.

Tech Stack: Java, Groovy, Grails, Spring Boot Micro-services, **OAuth**, TDD, **SonarQube**

WesBank | 2018 – 2020

- Delivered 56+ Spring Boot microservices into production.
- Led adoption of TDD, BDD, **SonarQube**, **Bitbucket** and Confluence.
- Designed a KYC Rules Engine using Drools and RabbitMQ.
- Architected integration services for LDAP, App Notifications, CT02, CIS, Actimize and Documentum.
- Produced solution designs, architectural artifacts and technical documentation.
- Received **Kubernetes training** from WesBank
- Completed **AWS** training through Udemy and online practical for Just over 1 Year .

Tech Stack: Java, **Spring Boot**, **SOAP**, **RabbitMQ**, **Docker**, Docker Compose, Portainer, **SonarQube**, **Postman**, JUnit, Mockito, TDD, BDD, **Drools**, **Kubernetes**, Elasticsearch, Logstash, Kibana, LDAP, IBM ODM, **Nexis**, **Bash**, **Java API**

Sanlam | 2020

- Developed Spring Boot microservices integrated with IBM ODM Business Rules Management Engine.
- Used IBM Rule Designer to develop process flow diagrams.

Tech Stack: Java, **Spring Boot**, JUnit, TDD, **IBM ODM**, **GitLab**

Momentum | 2017 – 2018

- Was the Full-stack Developer Team Lead
- Responsible for architecture, TDD adoption, code quality.
- Worked with **Docker**, **Docker Swarm** and **Ansible** automation

Tech Stack: Java, **Angular 2**, **Typescript**, **Bootstrap**, **Docker**, Docker Swarm, Postman, JUnit, Mockito, TDD, IBM ODM, **Ansible**, **GitLab**

FleetDomain

Solutions Architect | Infrastructure Architect | Senior Java Developer | May 2016 – Jan 2017

- Implemented enterprise security policies, disaster recovery, backup automation and UAT environments.
- Designed High Availability production architecture and hot backup solutions.
- Reverse-engineered and optimised Groovy/Grails enterprise applications.
- Modernised deployment architecture and server provisioning processes.

Tech Stack: Java, Groovy, Grails, **Ubuntu Server**, Windows Server, **Linux Administration**, Disaster Recovery, Security Architecture, Backup Automation, **Bash**, **GIT**

Standard Bank

Senior Java Developer | Jul 2015 – May 2016

- Developed enterprise OTP authentication services for Africa and Offshore Banking platforms.
- Delivered JEE8 interceptor framework and Vasco OTP integration.
- Applied TDD and BDD using JUnit, Mockito, WireMock, JBehave and Serenity.

Tech Stack: Java, JEE8, **SOAP**, **Maven**, JUnit, Mockito, WireMock, JBehave, Serenity BDD, SOAP, Web Services. **Centos Linux**, Puppet, **Java API**.

Cell C

Java Solution Architect | Infrastructure Architect | Senior Java Developer | Mar 2014 – Jul 2015

- Redesigned enterprise systems for scalability and performance.
- Changed the Architecture of Badly performing system
- Analyzed the Siemens Nokia Synix architecture Produced UML models.
- Optimized the Complete System Design.
- Made substantial improvement to performance

Tech Stack: Java, UML, Enterprise Architect, EJB, Oracle SQL, **Centos Linux**, Percona MySQL, JUnit, TDD, Javascript, **Bash**.

Signio

Senior Java Developer | Nov 2009 – Feb 2014

- Developed Standard Bank integration gateways and insurance platform integrations.
- Delivered Jasper Reports solutions and Groovy/Grails applications.
- Developed a project for the VAF5 System to send requests to the NATIS gateway
- Delivered new gateways to the Insurance companies

Tech Stack: Java, EJB3, SOAP, **Centos Linux**, Percona MySQL, Groovy, Grails, **Jasper Reports**, **Bash**

Paracon

ESKOM

Infrastructure Architect | **Senior Java Developer** | Apr 2007 – Jul 2009

- Delivered architecture and integration designs for the Online Vending System
- Installed and configured enterprise integration platforms and middleware.
- Developed BPEL Business Processes for SOAP integration connecting SAP using Sun Java CAPS.
- Worked as **Infrastructure Architect** for 2 years.
- We used the **VETO and VETRO SOAP Integration Patterns**

Tech Stack: Java, **SOAP**, **BPEL**, **LDAP**, Sun Java CAPS, Infrastructure Architecture, **Ubuntu Linux Server**, **Solaris Unix**, **Bash**, **VMware ESX Server**, **SUN VM Server**

ABSA Bank

Senior Java Developer | Jan 2006 – Mar 2007

- Developed Home Loans portal solutions using JSR168 Portlets and JSF.
- Participated in delivery of the production Home Loans platform.

Tech Stack: Java, JSR168 Portlets, JSF, **IBM AIX Unix**

Standard Bank

Senior Java Developer | Jul 2002 – Dec 2005

- Developed Home Loans portal solutions using JSR168 Portlets and JSF.
- Participated in delivery of the production Home Loans platform.

Tech Stack: Java, Java EJB2, Vignette CMS, **Solaris Unix**

IBM Global Services

Infrastructure Architect | Java Developer | Feb 1998 – Jun 2002

- Developed components of the National Online Casino Gambling Monitoring System.
- Worked as Infrastructure Architect.
- Delivered **Infrastructure Architecture for StanLib project**
- Created Architectural artifacts for the GJMC project for Johannesburg Metro Council
- I conducted the **CMM Maturity Rating of Rubico** for the World Bank
- Was part of the **IBM GS Method Methodology** training team of IBM

Tech Stack: Java, Java EJB, Java J2EE, RUP, Visual Age for Java, UML Sybase SQL server, IBM DB2, Windows NT Server, O/S2

Education, Professional Training & Qualifications

National Higher Diploma – Production Management

Tshwane University of Technology | Graduation 1989

National Diploma – Electrical Engineering L/C

University of Johannesburg | Graduation 1980

Matric

TED | 1975

Professional Training:

SQL, Java, EJB JEE, UML, C++, Kubernetes, Docker, Jenkins, Terraform, Azure, AWS, Metasploit and Agile Methodologies.

Training providers include **IBM, Damelin and Udemy**

Publications, Open Source & Community Contributions

Published Microservices White Paper on Test Driven Development.

- <https://www.dvt.co.za/white-paper-building-cloud-ready-microservices-using-industry-best-practices>

Published 50 Open Source projects on GitHub.

- Created the K3S Kubernetes Auto Cluster Generator
- The Claude Code Multi Agent Development project With Policies.

- Technical training in Java, Kubernetes, DevOps and AI-Assisted Development.

Published Videos in DevOps YouTube channel

- Videos covering Java, Enterprise Java, **Quarkus**, TDD, **Docker**, **Kubernetes**, Portainer, **Proxmox**, **Terraform**/OpenTofu, **Helm**, **Jenkins**, **Ansible**, AI Agentic Development and N8N.

Professional Memberships

- SAIETE – South African Institute for Technician Engineers
- SAIEE – South African Institute for Electrical Engineers
- JUGSA – Java User Group of South Africa
- GLUG – Gauteng Linux User Group

References

Professional and personal references available upon request.